

Sample test

Name: _____ **Date:** _____

Multiple Choice questions: Each question is worth 1 mark

1. As a mine is drilled deeper and deeper into the Earth, conditions become:
 - a) Cooler
 - b) Wetter
 - c) Hotter
 - d) Smellier
2. Which of the following is a list of the three rock types?
 - a) igneous, sedimentary, magma
 - b) igneous, hydrous, metamorphic
 - c) sedimentary, metamorphic, sediments
 - d) igneous, sedimentary, metamorphic
3. Rocks deep down in the mantle are very hot, but they are not totally melted because:
 - a) The temperature is still too low for rocks to melt
 - b) They are also under very high pressure
 - c) They have a different crystal structure
 - d) They don't contain any water
4. Sandstone is a rock that contains small grains of sand that are cemented together. These grains of sand probably came from:
 - a) Other rocks that were broken down
 - b) A volcano
 - c) Crystals that formed in the ocean
 - d) Meteorites and asteroids that have hit the Earth
5. Which one of the following is NOT a mineral?
 - a) Quartz
 - b) Granite
 - c) Biotite
 - d) Feldspar

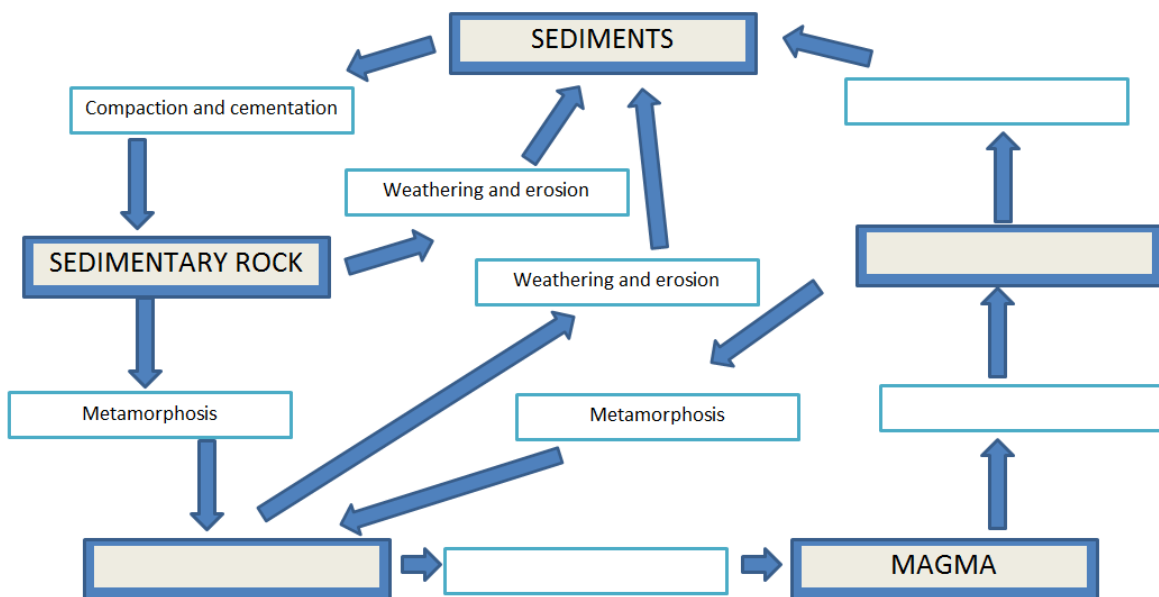
6. A rock contains large crystals that appear to have grown together and are tightly interlinked. The rock is most likely;
 - a) A volcanic igneous rock
 - b) A biogenic rock
 - c) A sedimentary rock
 - d) A plutonic igneous rock
7. The Australian landscape has many granite boulders. Granites are rocks that form deep beneath the Earth. They are lying on the surface today because;
 - a) Volcanoes have blasted them to the surface.
 - b) Weathering and erosion have worn away the rocks above them.
 - c) Glaciers have carried them from a long distance away.
 - d) They are left over from mining activities.
8. A sedimentary rock made from pebbles and gravel cemented together is;
 - a) Sandstone
 - b) Mudstone
 - c) Conglomerate
 - d) Limestone
9. The sedimentary rock in question 8 is likely to have formed from sediments from:
 - a) A fast flowing stream or river
 - b) A river estuary (where the river meets the sea or ocean)
 - c) The shallow ocean
 - d) A desert
10. Metamorphic rocks are usually created;
 - a) During a volcanic eruption
 - b) As rocks experience high temperatures and pressures deep underground
 - c) At the bottom of very deep oceans
 - d) In glaciers
11. The study of large beds of sedimentary rocks has shown that:
 - a) The sediments are laid down horizontally with the oldest sediments at the bottom
 - b) The sediments are laid down horizontally with the youngest sediments at the bottom
 - c) The sediments are laid down vertically with the oldest sediments at the left
 - d) The sediments are laid down vertically with the youngest sediments at the bottom

12. The rock cycle describes how;
- New rocks can form from molten rock that solidifies
 - New rocks can form as rocks experience extreme temperatures and pressures
 - New rocks can form as sediments are compacted deeper and deeper under other sediments
 - All of the above
13. The most important, and profitable, mineral mined in Australia is;
- diamond
 - gold
 - opal
 - haematite
14. A rock that contains fossils is most probably;
- An igneous rock
 - A metamorphic rock
 - A sedimentary rock
 - An igneous or a metamorphic rock

Short Answer Questions (Marks are indicated on each question)

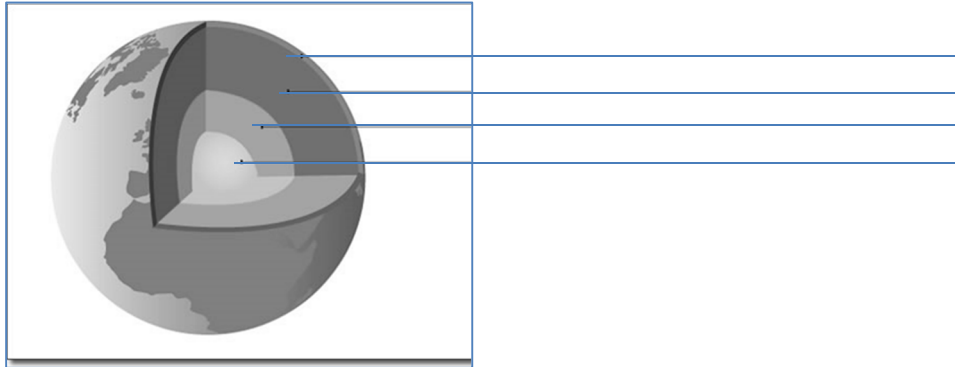
- 15 Fill in the empty spaces on the rock cycle diagram using these terms. **(5 marks)**

igneous rocks, metamorphic rocks, melting, weathering and erosion, solidification



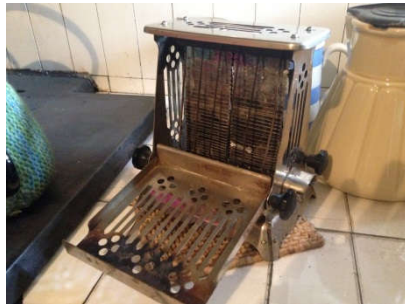
- 16 Label the four layers indicated on this diagram.

(4 marks)



- 17 Toasters often have a layer of a mineral to support the element. This mineral is used because it is extremely heat resistant. In appearance it has flat sheets that can be flaked off, is quite soft, and has a silvery colour.

(3 marks)



Use the information provided in the tables at the end of this test to identify this mineral.

Answer:

- 18 The rock cycle describes how rocks break down and how new rocks are made. Which part of the rock cycle has happened the most in Australia during recent geological times?

Explain your answer:

(2 marks)

- 19 Place each of these rocks in the appropriate column in the following table. **(3 marks)**

mudstone, basalt, marble, slate, granite, conglomerate

ROCK TYPES

Igneous	Sedimentary	Metamorphic

- 20 Much of the Australian outback has a rusty, red colour. Explain why this is important for Australia's export industries and economy. **(2 marks)**

- 21 Complete the table by matching each of these four minerals with the product made from them. **(4 marks)**

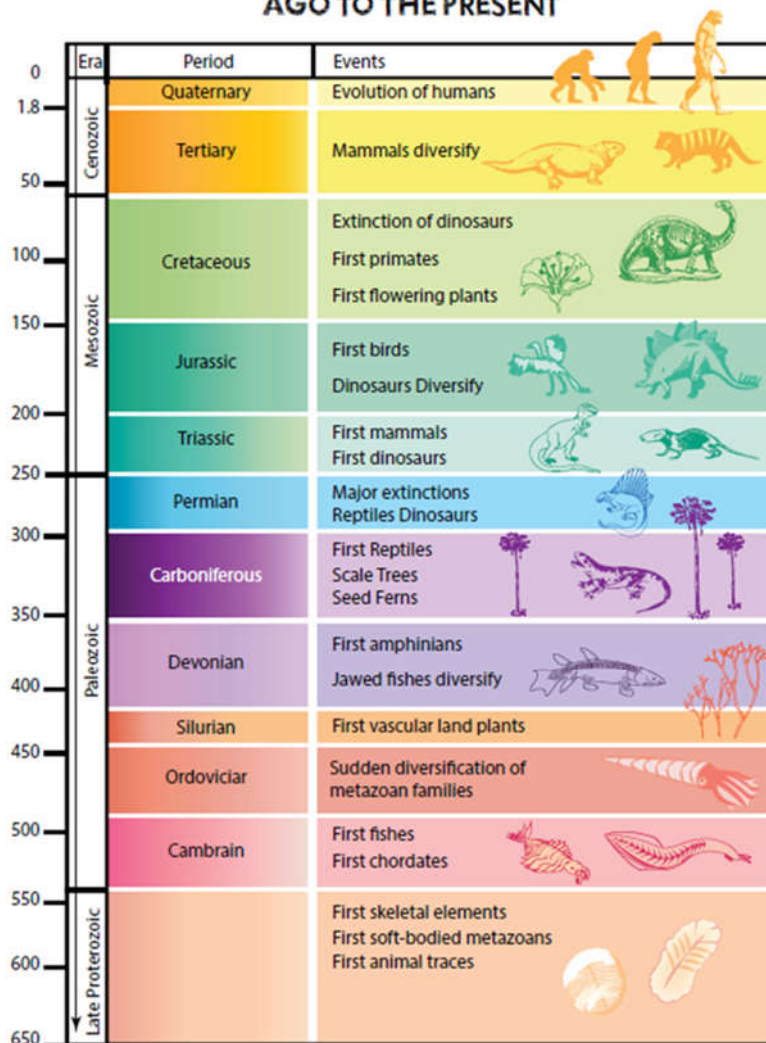
bauxite, halite, calcite, haematite

MINERAL	PRODUCT
	cement
	table salt
	steel, iron
	aluminium

22. Ichthyosaurs are large marine reptiles that first appeared in the fossil record in the early Triassic period. They became extinct in the late Cretaceous period. **(2 marks)**



GEOLOGIC TIME SCALE, 650 MILLION YEARS AGO TO THE PRESENT



- a) How many years did this group live on Earth?

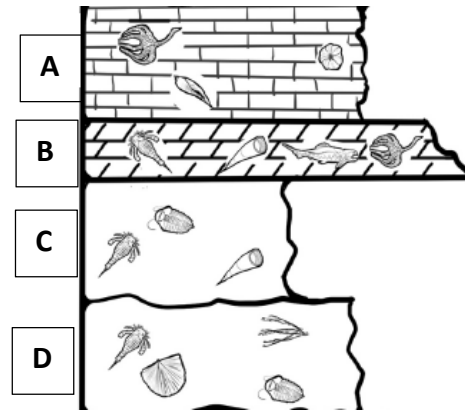
- b) In what era did the Ichthyosaurs live?

23. Geologists studied three roadside cuttings to look at the sediments and the fossils they contained to try to work out the stratigraphic order of the sediments from oldest to youngest.

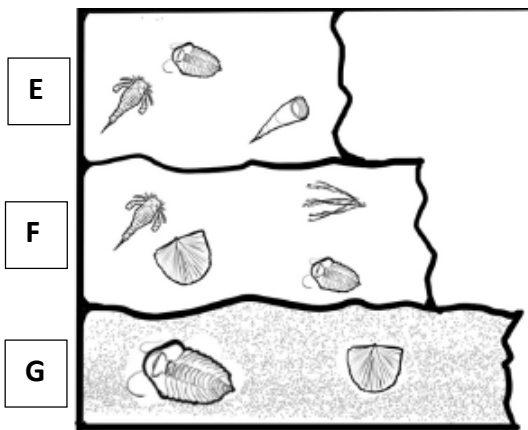
Key

	Sandstone
	Shale
	Limestone
	Dolomite

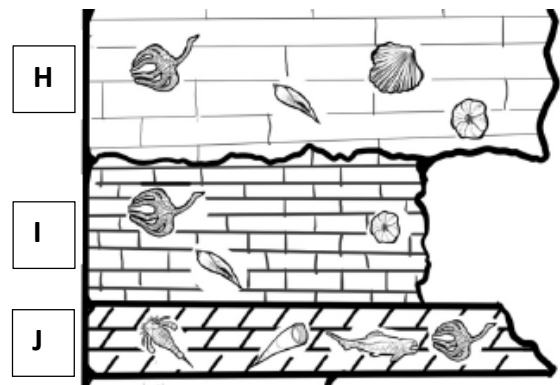
Cutting 1



Cutting 2

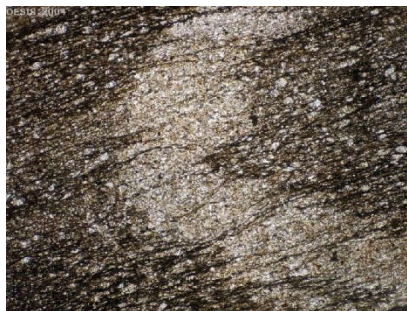
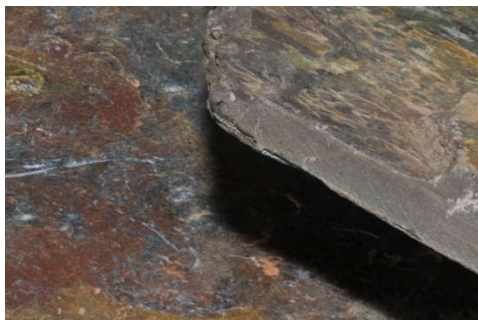


Cutting 3



- Identify two sediments from sediments A – J that contain exactly the same fossils.
(1 mark)
- In Cutting 1 which is the oldest sediment? (1 mark)
- Out of all the sediments A – J, which is the oldest sediment? (2 marks)
- Out of all the sediments A – J, which is the youngest sediment? (2 marks)

24. This is a picture of a rock that is sometimes used in building work. It has a layered structure.



A close examination of the tiny crystals it contains (right hand image) suggests that it is made of 4 different minerals with the following characteristics. **(5 marks)**

Mineral 1	Hardness	7
	Colour	colourless
	Lustre	glassy
	Cleavage	no



Mineral 2	Hardness	2.5
	Colour	silvery
	Lustre	glassy
	Cleavage	nice flat sheets



Mineral 3	Hardness	3
	Lustre	Glassy
	Colour	black
	Cleavage	Nice flat sheets



Mineral 4	Hardness	6
	Colour	Black, silver grey or dull red
	Lustre	metallic
	Cleavage	no



Use the information provided in the following tables to identify the four minerals and the type of rock.

Mineral 1 -

Mineral 2 -

Mineral 3 -

Mineral 4 -

Name of rock -

Some common rocks and the minerals they contain.

Rock	Minerals
Andesite	plagioclase feldspar, augite, hornblende
Slate	muscovite, quartz, hematite, biotite
Granite	quartz, biotite, potassium feldspar, plagioclase feldspar
Limestone	mostly calcite
Granodiorite	quartz, biotite, potassium feldspar, plagioclase feldspar
Sandstone	mainly quartz sand grains held together in a silica cement

Some common rock forming minerals

Mineral	Hardness	Cleavage	Streak	Colour	Lustre
Quartz	7	No	white	colourless through to black	glassy, but large pieces may appear dull
Biotite mica	2.5 - 3	Yes, flat sheets	white	dark brown, greenish-brown, blackish-brown	glassy or pearly
Muscovite mica	2 - 2.5 (when split along sheets)	Yes, flat sheets	white	white, grey, silvery	glassy, silky or pearly
Feldspar (Orthoclase)	6 - 6.5	Yes	white	colourless, greenish, greyish yellow, white, pink	glassy or pearly
Feldspar (Plagioclase)	6 - 6.5	Yes	white	white, grey, bluish white, reddish white, greenish white	glassy
Calcite	3	Perfect cleavage	white	colourless or white, also grey, yellow, green	glassy or pearly
Hornblende	5 - 6	Not consistent	pale grey	black, dark green	glassy or dull
Haematite	5.5 - 6.5	No	Black, silver grey, bright red to dark red	metallic grey, dull to bright red	metallic